

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of MSc Physics Examination April 2018

PS951 Molecular Physics

Date: 03-05-2018, Day: Wednesday Time: 10.00 AM To 10.20 AM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q-1 Choose the correct answer for the following questions [10]

1. Nuclear spin or intrinsic spin angular momentum will be zero for 1
 - a. ^{12}C
 - b. ^{10}B
 - c. ^2H
 - d. ^{19}F
2. Which is incorrect? 1
 - a. Heteronuclear diatomic molecules exhibit rotational spectra
 - b. CO_2 and CH_4 show pure rotational spectra
 - c. Isotope effect is observed in vibrational as well as rotational spectra
 - d. The electronic spectra are observed for the heteronuclear as well as for homonuclear diatomic molecules.
3. Pure rotational spectra is observed in region and electronic spectra is 1
observed in region of electromagnetic spectrum.
 - a. Visible, microwave
 - b. Ultraviolet, visible
 - c. Microwave, Ultraviolet
 - d. Infra-red, microwave

4. In a rotational transition $J = 4 \rightarrow J = 5$, the wavenumber of the absorbed radiation would be 1
 - a. $10B$
 - b. $8B$
 - c. $2B$
 - d. $4B$
5. Dimension of μ_B (Bohr magneton) is 1
 - a. $J s^{-1}$
 - b. $J \cdot s$
 - c. $J T^{-1}$
 - d. $J \cdot C s^{-1}$
6. Which of the following will not show IR spectrum but will give Raman spectrum? 1
 - a. HCl
 - b. N_2
 - c. CO
 - d. CN
7. For AlH, moment of Inertia is $4.44 \times 10^{-47} \text{ kg m}^2$ and angular velocity at $J=15$ is $3.68 \times 10^{13} \text{ rad/s}$. The angular momentum would be J·s 1
 - a. 1.63×10^{33}
 - b. 1.21×10^{-34}
 - c. 1.63×10^{-33}
 - d. 1.36×10^{-34}
8. Among which of the following molecule, rotational spectra will be observed? 1
 - a. CO_2
 - b. CO
 - c. CH_4
 - d. O_2
9. If ν is vibrational quantum number, the distributions of the probability density $|\Psi_\nu|^2$ will be 1
 - a. maximum at midpoint for $\nu = 1$
 - b. maximum for $\nu = 0$ at mid-point
 - c. minimum at turning points for $\nu = 2$
 - d. None of the above
10. An NMR signal for a compound is found to be 180 Hz downward from TMS peak using a spectrometer operating at 60 MHz. The chemical shift is 1
 - a. 3 ppm
 - b. 60 ppm
 - c. 6 ppm
 - d. 18 ppm

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Date: 03-05-2018, Day: Thursday Time: 10.20 AM To 12.00 PM Maximum Marks: 25

Instructions:

1. **Section I and II must be attempted in SINGLE ANSWER SHEET.**
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION –I

Q - II Answer the following questions as directed **10**

1. The force constant of the bond in CO molecule is 1870 Nm^{-1} . Calculate the frequency of vibration of the molecule and the spacing between its vibrational energy levels in eV. Given that the reduced mass of CO = $1.14 \times 10^{-26} \text{ kg}$. 2
2. Discuss the rotational spectrum of a diatomic molecule, treated as a non-rigid rotator. 2
3. What are the discrepancies in classical explanation of Raman effect? 2
4. Explain the principle of ESR. 2
5. Calculate the strength of the magnetic field to give a precessional frequency of 100 MHz for ^{17}O nucleus. ($g_N = -0.757$; and $I = 5/2$ for ^{17}O ; $\mu_N = 5.05 \times 10^{-27} \text{ JT}^{-1}$) 2

SECTION - II

Q-III Answer the following questions as directed (any five) **15**

1. Give an account of the salient features of near infra-red spectrum of a diatomic molecule like HCl. Briefly state how these observations are accounted for quantum mechanically. Why are such spectra not obtained for molecules having identical nuclei? 3
2. Discuss the variation in intensity distribution in electronic spectra of different systems (O_2 , CO and I_2). Explain with examples by showing the spectra and state the principle responsible for such variation. 3
3. The far infra-red spectrum of H^1Br^{79} consists of a series of lines spaced 17 cm^{-1} apart. Find the internuclear distance of H^1Br^{79} . 3
4. Discuss the ESR spectrum of Hydrogen atom. What will be the energy (corrected to first order) and corresponding allowed transitions in applied magnetic field B. 3
5. (i) What is screening constant σ in NMR? 3
(ii) Why TMS is usually selected as reference material in NMR?
(iii) In the NMR spectrum of ^{14}N with $I=1$, how many spectral lines will be observed?
6. Write note on: IR spectroscopy versus Raman spectroscopy. 3